serviceOrchestrationPushManagement

Service Description

Service ID: *"serviceOrchestrationPushManagement"*

Abstract
This document describes the `serviceOrchestrationPushManagement` service, version 1.0.0, produced by the DynamicServiceOrchestration core system of Arrowhead 5.2.2.

Contents

1 Overview **3**
1.1 Significant Prior Art . . . 3
1.2 How This Service Is Meant to Be Used . . . 3
1.3 Important Delimitations and Dependencies . . . 3

2 Service Interface **4**
2.1 operation subscribe . . . 4
2.2 operation unsubscribe . . . 4
2.3 operation trigger . . . 4
2.4 operation query . . . 4

3 Information Model **5**
3.1 struct MetadataRequirement . . . 5
3.2 struct OrchestrationJob . . . 5
3.3 struct OrchestrationNotifyInterface . . . 5
3.4 struct OrchestrationPushJobListResponse . . . 5
3.5 struct OrchestrationPushTrigger . . . 5
3.6 struct OrchestrationRequest . . . 6
3.7 struct OrchestrationServiceRequirement . . . 6
3.8 struct OrchestrationSubscriptionListRequest . . . 6
3.9 struct OrchestrationSubscriptionListResponse . . . 6
3.10 struct OrchestrationSubscriptionQueryRequest . . . 7
3.11 struct OrchestrationSubscriptionRequest . . . 7
3.12 struct OrchestrationSubscriptionResponse . . . 7
3.13 struct Page . . . 7
3.14 struct QoSRequirement . . . 8
3.15 Primitives . . . 8

4 References **9**

5 Revision History **10**
5.1 Amendments . . . 10
5.2 Quality Assurance . . . 10

1 Overview

This document describes the `serviceOrchestrationPushManagement` service, which provides operator management of push orchestration: subscriptions made on behalf of consumers, and triggering of push orchestration. It is produced by the DynamicServiceOrchestration system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

1.1 Significant Prior Art

Corresponds to the Arrowhead 4.x orchestration store push features.

1.2 How This Service Is Meant to Be Used

Operators and support systems (e.g. the Choreographer, PlantDescriptionEngine) create subscriptions for target systems, query and remove them, and trigger the execution of subscriptions; the orchestrator then sends the results to the subscribers' notify interfaces.

1.3 Important Delimitations and Dependencies

Push notifications are sent by worker threads (`push.orchestration.max.thread`); delivery is best effort and its outcome is recorded in the orchestration history.

2 Service Interface

This section describes the interfaces to the `serviceOrchestrationPushManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

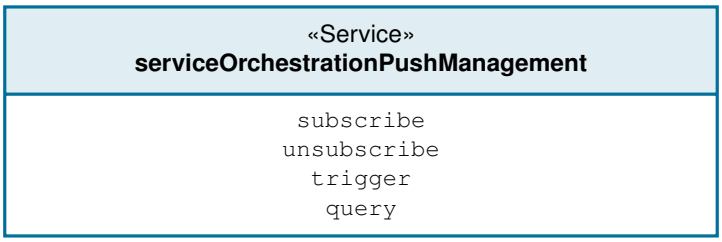


Figure 1: SysML-style block of the `serviceOrchestrationPushManagement` service and its operations.

The following interface operations are available.

2.1 operation subscribe (OrchestrationSubscriptionListRequest) : OrchestrationSubscriptionListResponse

Creates subscriptions on behalf of target systems.

2.2 operation unsubscribe (List<Identifier>)

Removes subscriptions by identifier.

2.3 operation trigger (OrchestrationPushTrigger) : OrchestrationPushJobListResponse

Starts push orchestration for the selected subscriptions (by target system or identifier); returns the started jobs.

2.4 operation query (OrchestrationSubscriptionQueryRequest) : OrchestrationSubscriptionListResponse

Returns a page of subscriptions matching the filter.

3 Information Model

Here, all data objects that can be part of the `serviceOrchestrationPushManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.15, which are used to represent things like hashes and identifiers.

3.1 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

3.2 struct OrchestrationJob

One orchestration job (pull or push).

Field	Type	Description
id	Identifier	Identifier (UUID) of the orchestration job.
status	String	Status of the job: `PENDING`, `IN_PROGRESS`, `DONE` or `ERROR`.
type	String	Orchestration type: `PULL` or `PUSH`.
requesterSystem	Name	Name of the system that requested the orchestration.
targetSystem	Name	Name of the consumer system the orchestration was made for.
serviceDefinition	Name	Service definition name.
subscriptionId	Identifier	Subscription that started the push job.
message	String	Status message, e.g. `1 local result: DataManager`.
createdAt	DateTime	UTC time of creation.
startedAt	DateTime	UTC time when the job started.
finishedAt	DateTime	UTC time when the job finished.

3.3 struct OrchestrationNotifyInterface

Notification endpoint of push orchestration.

Field	Type	Description
protocol	String	Protocol of the notification: `HTTP`, `HTTPS`, `MQTT` or `MQTTS`.
properties	Map<String, String>	For HTTP(S) `address`, `port`, `method`, `path`; for MQTT(S) `topic` (and broker `address`, `port`).

3.4 struct OrchestrationPushJobListResponse

Push jobs started by a trigger.

Field	Type	Description
jobs	List<OrchestrationJob>	Orchestration jobs started by the trigger.

3.5 struct OrchestrationPushTrigger

Selects the subscriptions to execute now.

Field	Type	Description
targetSystems	List<Name>	Consumers whose subscriptions are executed.
subscriptionIds	List<Identifier>	Subscriptions to execute.

3.6 struct OrchestrationRequest

An orchestration request.

Field	Type	Description
serviceRequirement	OrchestrationServiceRequirement	Requirements on the service the consumer wants to use.
orchestrationFlags	Map<String, Boolean>	Orchestration flags, e.g. `MATCHMAKING`, `ONLY_PREFERRED`, `ALLOW_INTERCLOUD`.
qualityRequirements	List<QoSRequirement>	Quality-of-service requirements (evaluated by DeviceQoSEvaluator).
exclusivityDuration	Number	Requested exclusive reservation of the provider in seconds.

3.7 struct OrchestrationServiceRequirement

Requirements on the requested service.

Field	Type	Description
serviceDefinition	Name	Service definition name.
operations	List<String>	Service operations the consumer wants to use.
versions	List<Version>	Accepted service versions.
alivesAt	DateTime	Only instances that are still registered (not expired) at this UTC time are matched.
metadataRequirements	List<MetadataRequirement>	Requirements on the service instance metadata; each map is one alternative.
interfaceTemplateNames	List<String>	Accepted interface templates, e.g. `generic_https`.
interfaceAddressTypes	List<String>	Address types the provider interface must offer.
interfaceProperty Requirements	List<MetadataRequirement>	Requirements on the interface properties; each map is one alternative.
securityPolicies	List<String>	Accepted interface security policies.
preferredProviders	List<String>	Provider systems to prefer.

3.8 struct OrchestrationSubscriptionListRequest

Subscriptions to create on behalf of consumers.

Field	Type	Description
subscriptions	List<OrchestrationSubscription Request>	Subscriptions to create.

3.9 struct OrchestrationSubscriptionListResponse

List of subscriptions.

Field	Type	Description
entries	List<OrchestrationSubscription Response>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.10 struct OrchestrationSubscriptionQueryRequest

Paged filter for subscriptions.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
ownerSystems	List<String>	Subscription owner names to match.
targetSystems	List<Name>	Target system names.
serviceDefinitions	List<Name>	Service definition names to match.

3.11 struct OrchestrationSubscriptionRequest

A push orchestration subscription.

Field	Type	Description
targetSystemName	Name	Consumer system; default the requester.
orchestrationRequest	OrchestrationRequest	The orchestration request executed for the subscription.
notifyInterface	OrchestrationNotifyInterface	Where and how the orchestration result is pushed to the target system.
duration	Number	Requested lifetime of the subscription in seconds; no value means unlimited.

3.12 struct OrchestrationSubscriptionResponse

A stored subscription.

Field	Type	Description
id	Identifier	Identifier (UUID) of the subscription.
ownerSystemName	Name	Name of the system that created the subscription.
targetSystemName	Name	Name of the consumer system the push orchestration is made for.
orchestrationRequest	OrchestrationRequest	The orchestration request executed for the subscription.
notifyInterface	OrchestrationNotifyInterface	Where and how the orchestration result is pushed to the target system.
expiredAt	DateTime	UTC time when the subscription expires.
createdAt	DateTime	UTC time of creation.

3.13 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

3.14 struct QoSRequirement

One quality-of-service requirement.

Field	Type	Description
type	String	Type of the QoS evaluation, as known by DeviceQoSEvaluator.
operation	String	QoS operation: `FILTER` or `SORT`.
requirements	Map<String, Object>	Parameters of the QoS requirement.

3.15 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

4 References

[1] J. Delsing, “DynamicServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, serviceorchestration-dynamic/documentation/v5.2.2/DynamicServiceOrchestration-v5.2.2-SysD.pdf.

[2] ——, “serviceOrchestrationPushManagement HTTP/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.2.2, 2026, dynamicServiceOrchestration-v5.2.2-serviceOrchestrationPushManagement-HTTP-IDD.pdf.

[3] ——, “serviceOrchestrationPushManagement MQTT/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.2.2, 2026, dynamicServiceOrchestration-v5.2.2-serviceOrchestrationPushManagement-MQTT-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	